

CS111 Reference Document

String Methods

Method Signature	Description
<code>int length()</code>	Provides the number of characters in the string
<code>char charAt(int index)</code>	Returns the character at an index
<code>int indexOf(String str)</code>	Returns the index of the given string
<code>String substring(int startIndex, int endIndex)</code>	Returns a substring between the starting and ending index, exclusive of end
<code>boolean equals(String str)</code>	Returns true if both strings are the same
<code>boolean equalsIgnoreCase(String str)</code>	Returns true if both strings are the same, case-insensitive
<code>int compareTo(String str)</code>	Compares two strings to each other, returning the difference in ASCII values
<code>int compareToIgnoreCase(String str)</code>	Compares two strings to each other, returning the difference in ASCII values, case-insensitive
<code>String[] split(String regex)</code>	Splits the string by a separator
<code>String strip()</code>	Removes whitespace from both ends
<code>char[] toCharArray()</code>	Converts the string into an array of characters

StringBuilder Methods

Method Signature	Description
<code>StringBuilder append(String s)</code>	Adds the string to the StringBuilder
<code>StringBuilder delete(int startIndex, int endIndex)</code>	Deletes characters between the indexes, exclusive of end index
<code>StringBuilder deleteCharAt(int index)</code>	Deletes the character at the given index
<code>String toString()</code>	Converts the StringBuilder into the String version

ArrayList/LinkedList Methods

Method Signature	Description
<code>E get(int index)</code>	Retrieves the value at the given index
<code>boolean add(E element)</code>	Adds the element to the List

Method Signature	Description
<code>void add(int index, E element)</code>	Adds the element at the index to the List
<code>boolean contains(E element)</code>	Checks if the element is in the list
<code>int indexOf(E element)</code>	Returns the index of an element
<code>E remove(int index)</code>	Removes the value at an index
<code>E set(int index, E element)</code>	Sets the value at an index to the element
<code>int size()</code>	Returns the number of elements in the list

HashSet Methods

Method Signature	Description
<code>boolean add(E element)</code>	Adds the element to the Set
<code>boolean contains(E element)</code>	Checks if the element is in the set
<code>boolean remove(E element)</code>	Removes the element
<code>int size()</code>	Returns the number of elements in the set

HashMap Methods

Method Signature	Description
<code>boolean containsKey(Object key)</code>	Returns whether the key is in the map
<code>V get(Object key)</code>	Returns the value at the key, null if not found
<code>boolean isEmpty()</code>	Returns whether the map is empty
<code>int size()</code>	Returns how many elements are in the map
<code>V put(K key, V value)</code>	Puts the key-value pair in the map, replaces if the value already exists
<code>V putIfAbsent(K key, V value)</code>	Puts the key-value pair in the map if-and-only-if not already present
<code>V getOrDefault(K key, Object defaultValue)</code>	Gets the value at a key or a default value if not present